

The empirical recurrence for OEIS A235829: recovering a conjecture that is not written down

Adrian Perez Fontelles
Independent researcher

7 September 2026

Abstract

OEIS A235829 records “Empirical recurrence of order 84”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 146$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 84, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 84 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture one cannot read

OEIS A235829 is “Number of $(n+1) \times (2+1)$ 0..2 arrays with the difference of the upper median and minimum value of each 2×2 subblock in lexicographically nondecreasing order columnwise and nonincreasing rowwise.”. It has offset 1 and begins

549, 8160, 110923, 1378463, 16385455, 187651928, 2100282123, 23113626964, ...

The entry states:

Empirical recurrence of order 84 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:51:32, revision 8) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 222$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 146$ of the 222 , and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 146$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 444$ exact terms of the model returns order 84 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{84} c_i a(n-i),$$

c_1	28	c_{22}	−42227304715742941	c_{43}	4876999584249679565798554	c_{64}	−78
c_2	−95	c_{23}	905344749193187668	c_{44}	−7879925422585902417793283	c_{65}	20
c_3	−4090	c_{24}	75929301673971525	c_{45}	−10736080543158206527505565	c_{66}	42
c_4	36050	c_{25}	−8270557313801264049	c_{46}	18629197316613598694230741	c_{67}	−3
c_5	229358	c_{26}	1978266191778889700	c_{47}	20579987004544815959668378	c_{68}	−19
c_6	−3572757	c_{27}	64317816187633067696	c_{48}	−38442623026446378199627791	c_{69}	27
c_7	−4061880	c_{28}	−31833033240923929420	c_{49}	−34234967236906160618754551	c_{70}	70
c_8	198909058	c_{29}	−429093923417614873566	c_{50}	69248863778046597897032934	c_{71}	−1
c_9	−223348960	c_{30}	301750021059632166584	c_{51}	49169932674463282601711699	c_{72}	−2
c_{10}	−7405721266	c_{31}	2470404166451028906755	c_{52}	−108799879307821999614314608	c_{73}	49
c_{11}	18919048373	c_{32}	−2167618702785633785601	c_{53}	−60510785911633038322049921	c_{74}	47
c_{12}	199374650912	c_{33}	−12329354792089191857967	c_{54}	148837297392065587567810684	c_{75}	−1
c_{13}	−742820073789	c_{34}	12662789192798481357106	c_{55}	63068691640802994101415440	c_{76}	−
c_{14}	−4046367619790	c_{35}	53522137969113932937303	c_{56}	−176815977599922146851491952	c_{77}	2
c_{15}	19852544249510	c_{36}	−62040597052436843105569	c_{57}	−54618832810329407572394268	c_{78}	9
c_{16}	63218238858128	c_{37}	−202573987193498164375209	c_{58}	181757673375304313839042308	c_{79}	−
c_{17}	−400538718339510	c_{38}	259177264196107377492835	c_{59}	37927545135937698917316212	c_{80}	−
c_{18}	−762289511978022	c_{39}	669486479962944455098065	c_{60}	−160909533241010988539002040	c_{81}	−
c_{19}	6404356039701485	c_{40}	−932350027575084613446986	c_{61}	−19424559473833458713972312	c_{82}	−
c_{20}	6911640961607482	c_{41}	−1933270189268154698723663	c_{62}	121949499510741773974720864	c_{83}	−
c_{21}	−83515534100656504	c_{42}	2906162564565783987804971	c_{63}	5264866184223125105066992	c_{84}	−

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 84, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $84 + 4$ terms removed returns order 84 as well, so $\deg q_{\min} = 84 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 15 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $84 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 84 that this sequence satisfies — and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A235829.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.